

BRIDGE ENGINEERING CHALLENGE

Let's Build Bridges!



WHAT IS A BRIDGE?

Structures that help us cross rivers, valleys, and roads

Types of Bridges We'll Learn!



Beam Bridges - flat and strong!



Arch Bridges - curved and beautiful!



Amazing examples from India!

BEAM BRIDGES

The simplest bridge: a flat surface supported at both ends!



ARCH BRIDGES

Strong curved structures that push weight to the sides



GATHER MATERIALS

Collect paper, straws,
tape, and cardboard
pieces.

DESIGN YOUR BRIDGE

Draw your bridge plan -
beam or arch style?

BUILD IT

Follow your design and
construct your bridge!

TEST ITS STRENGTH

Place coins on top - how
many can it hold?

MATERIALS NEEDED

Popsicle sticks or cardboard strips

Tape or glue

Small weights for testing

Paper for planning

YOUR CHALLENGE!

Build a bridge that can hold the most weight using only the materials provided!

Think Like an Engineer!

You just built your own bridge! Now it's time to think like a real engineer. Look at your bridge carefully. Did you make a beam bridge with a flat surface? Or did you create an arch bridge with a curved shape? Engineers always ask questions about their designs to make them even better!

WHICH BRIDGE TYPE DID YOU CHOOSE? WHY DO YOU THINK IT WORKED?

**GREAT JOB, YOUNG
ENGINEERS!**

You learned about beam and arch bridges today!